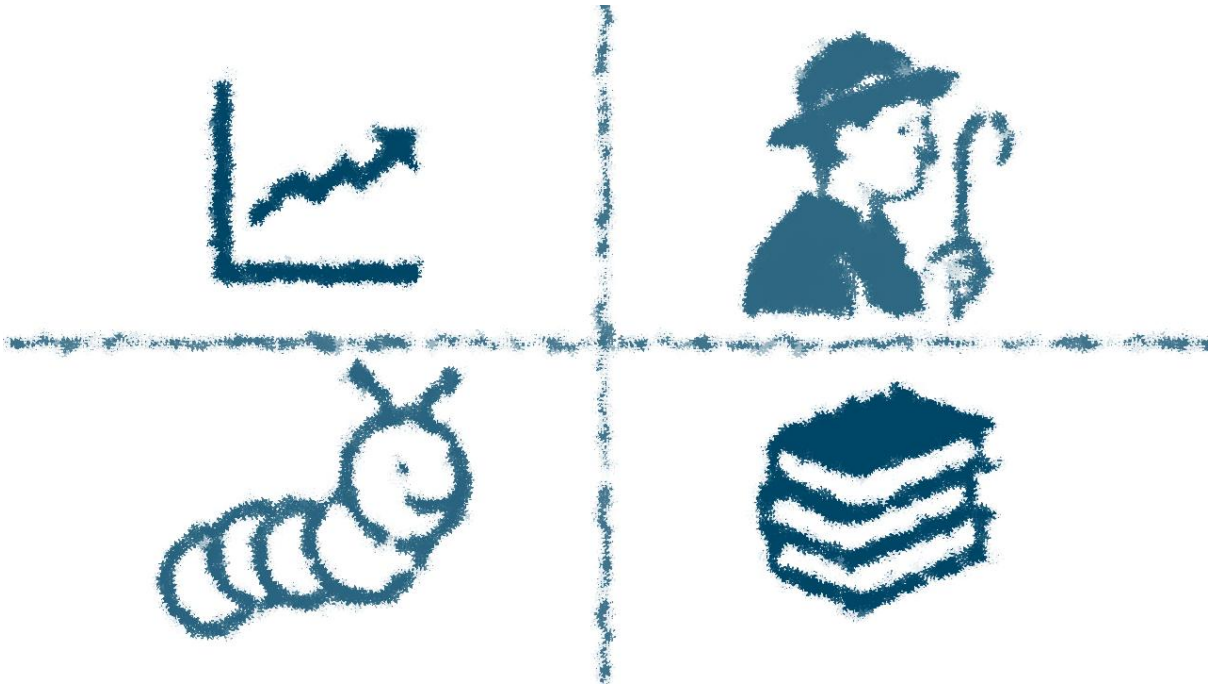
Topic project 9.1: Using time series analysis for sales and demand forecasting



CAM_DS_301 APPLYING ADVANCED DATA SCIENCE TECHNIQUES

F. ERATAM

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Introduction

Nielsen BookScan is a tracking service that collects real-time retail sales data for the book publishing industry. The historical sales data aims to support publishers in making data-driven decisions with regards to reordering, restocking, reprinting, and investing in new publications.

The current study incorporates two datasets provided by Nielsen to identify historical trends, by targeting the weekly sales volume recorded for the UK. The main analysis focuses on two specific books, The Alchemist and The Very Hungry Caterpillar, whereby a sales forecast is made using various classical and machine learning (ML) techniques including Autoregressive Integrated Moving Average (ARIMA) [1], Extreme Gradient Boosting (XGBoost) [2] and Long Short-Term Memory (LSTM) network[3].

Data Preprocessing

The dataset was resampled to ensure missing weeks were filled with zeros to represent 'no' sales.

ISBNs with sales data beyond 2024-07-01 were selected to study general patterns across different time periods, specifically comparing 1-12 and 12-24 years.

Based on the pattern analysis, details of which are presented in Appendix A, only the sales data from 2012 onwards were maintained for modelling the sales forecast of the two selected books.

Preliminary Analysis

The weekly timeseries data of the two books are presented in Figure 1, where the last 32 weeks have been reserved for forecasting. For both books, the annual occurrence of sharp peaks over a non-constant baseline combined with varying peak heights indicates the presence of both a strong seasonal and a non-linear long-term trend, which are confirmed by the decomposition plots in Figure 2.

Also, present are three 'no sales' gaps totalling 27 weeks which coincide with the UK-wide Covid lockdowns and distort the sales pattern significantly within the 2020-2022 window (with 2020 being the onset and 2021-2022, the recovery period). Since the zero values stem not from a lack of desire to purchase but an inability to purchase, the time series becomes mildly zero-inflated and thereby, traditional imputation techniques do not apply.

While the inclusion of zero values has implications for the forecasting models employed, inventing data, especially given the size of the disruption window is far from ideal. Therefore, an emphasis will be placed on observing and understanding the impact of a non-random event on the performance of the forecasting models. Note that the Augmented Dickey-Fuller test performed on both datasets, rejects the presence of a unit root and indirectly implies the Covid-19 disruption presents a short-term shock.

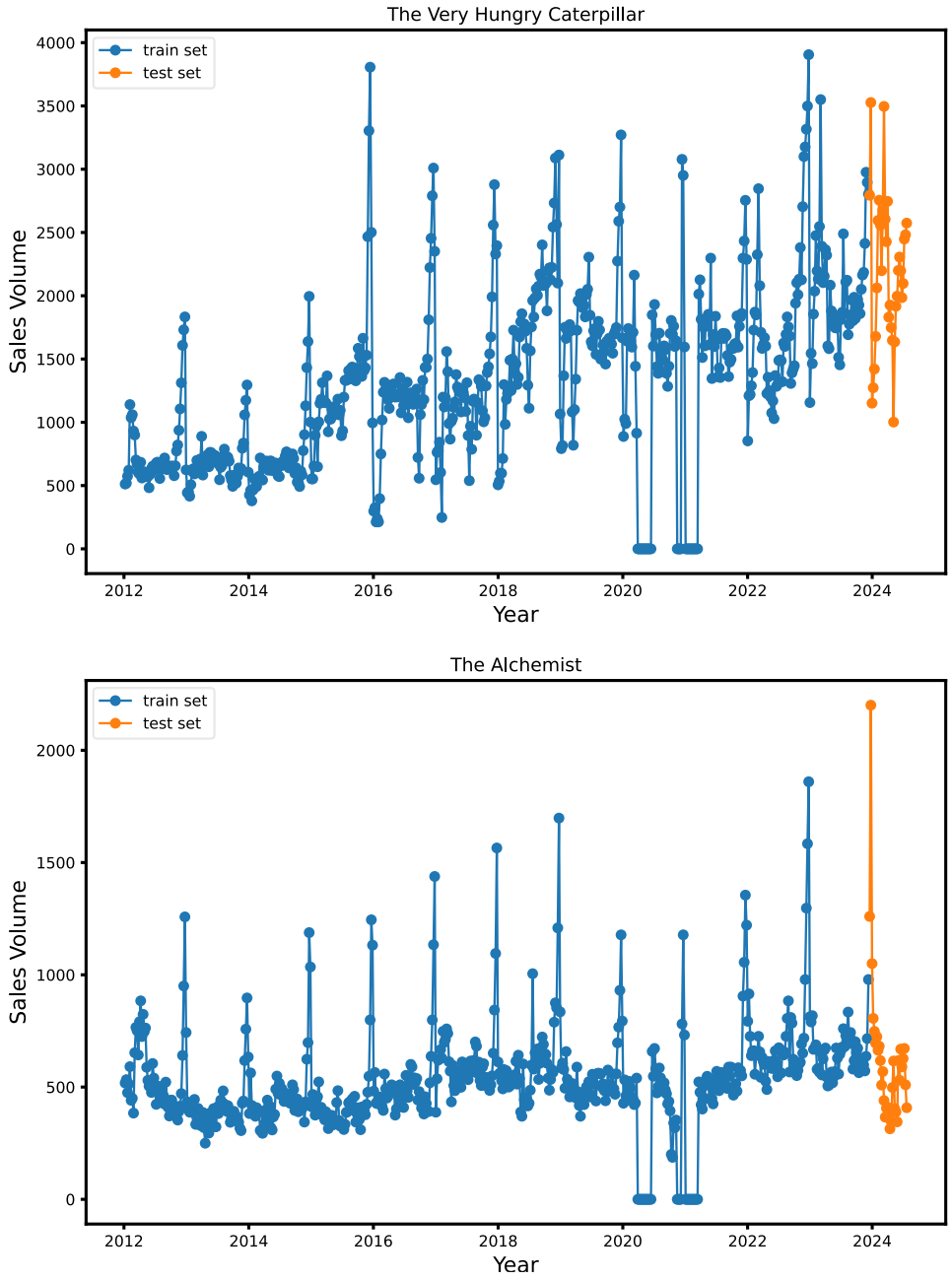


Figure 1: The weekly timeseries plots of the two books selected, 'The Very Hungry Caterpillar' (top) and 'The Alchemist' (bottom), showing the variation in sales volume in the UK within the early 2012 and late 2024 period. Note the data has been split into train-test sets with the last 32 weeks reserved for forecasting. Both timeseries exhibit an annual seasonality as indicated by the presence of sharp peaks occurring per year, as well as a non-linear long-term trend as indicated by the varying baseline. The two components along with non-constant variance indicate the two timeseries are non-stationary. In addition, 3 gaps totalling 27 weeks are present within the 2020-2022 window, which coincide with the UK Covid-19 lockdowns and indicate 'no sales' within that period. Notably, the sales volume recovers briefly with the ease of lockdown rules but the positive trajectory is once again interrupted by another lockdown period. The presence of zero values that are not true zeros but still caused by a non-random event, introduces complications for both the ARIMA and the ML models.

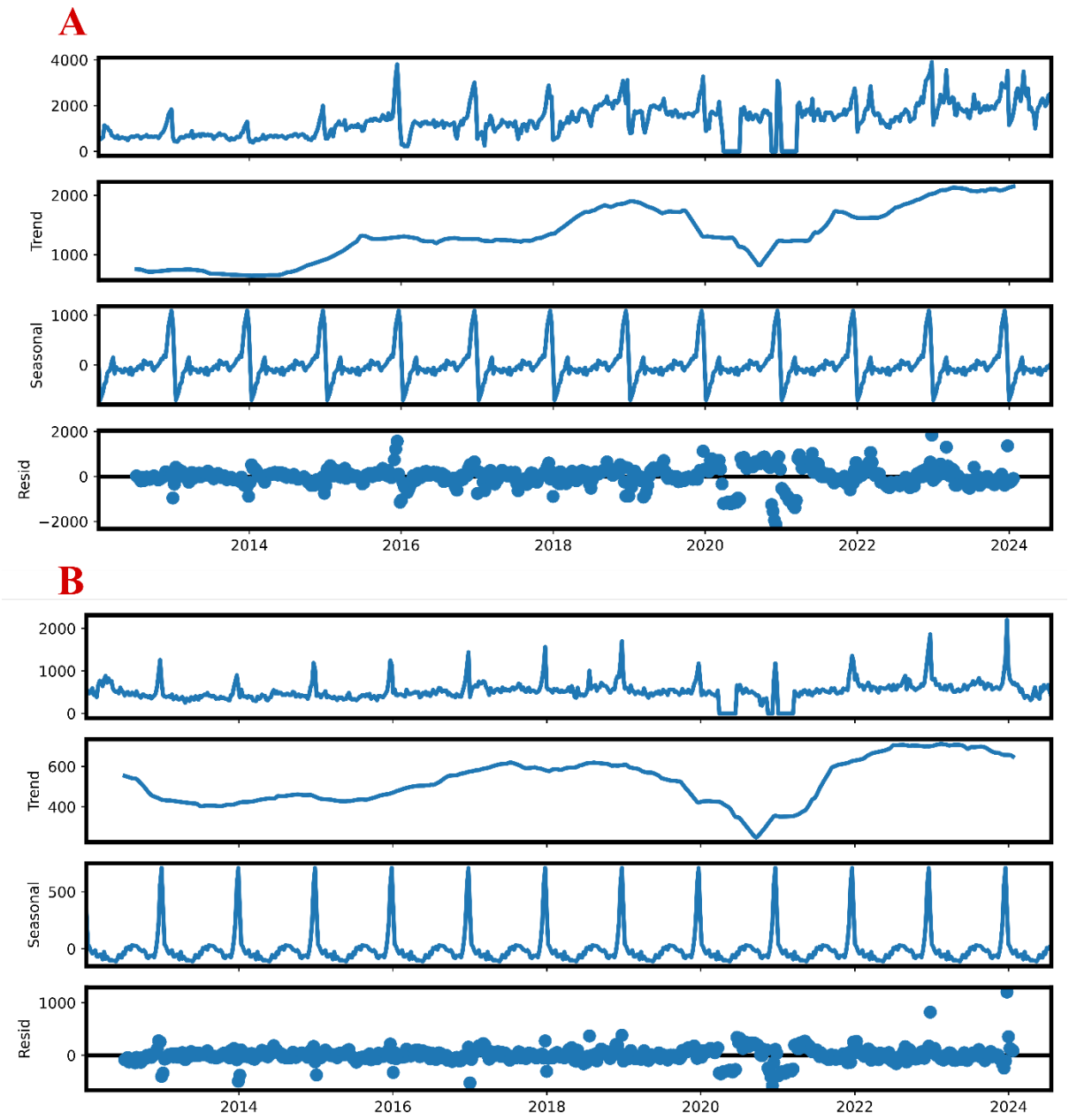


Figure 2: Classical decomposition plots obtained for 'The Very Hungry Caterpillar' (A) and 'The Alchemist' (B), depicting the fundamental aspects of a timeseries: the 'trend', 'seasonality', and residuals. In both cases, the long-term trend is disrupted by the period coinciding with the Covid-19 lockdowns. The positive trajectory which continually increases from 2014, albeit at different slower rates, suddenly dips within the 2020-2021 window before slowly recovering again. The sudden distortion causes the trend to appear quadratic for both books. The relatively complex seasonal patterns on the other hand follow an annual cycle. 'The Very Hungry Caterpillar' presents with a square peak form comprising a spike at the trailing edge, whereas 'The Alchemist' presents with a small peak immediately followed by a fast-decaying spike. Note that to account for the zero values, an additive seasonality was assumed. Finally, the residuals in both cases exhibit a very wide range and oscillate about a zero mean, with some structure remaining.

ARIMA Forecast

An auto ARIMA search was performed on both datasets to identify the best models which are described in Table 1. The results of the fit are shown in Figure 3, whereas an analysis of the residuals is provided in Appendix B.

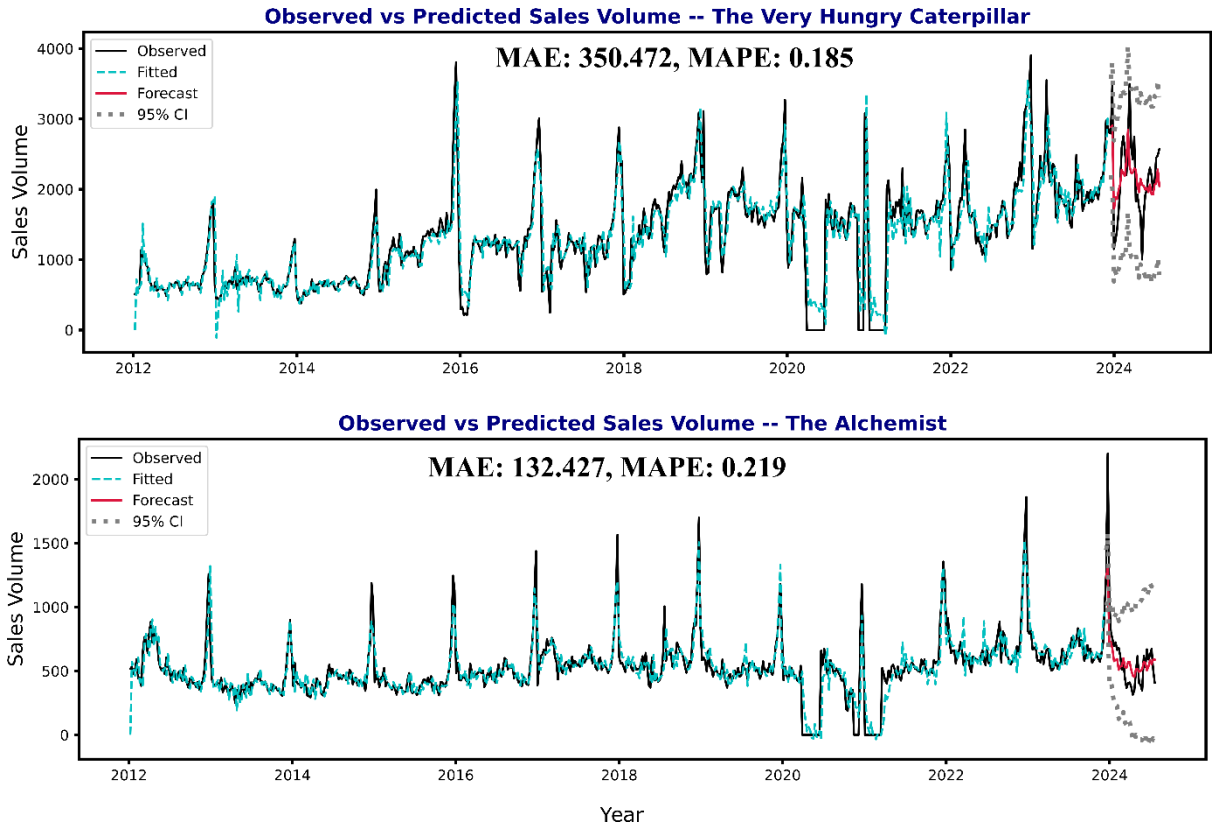


Figure 3: The SARIMA models fitted to 'The Very Hungry Caterpillar' (top) and 'The Alchemist' (bottom) timeseries and used to predict the forecast on the last 32 weeks. The fitted values are plotted in cyan, the predicted forecasts are shown in crimson and the confidence intervals (significance level, 0.05) are depicted as dotted grey lines. Also provided are the mean absolute (MAE) and mean absolute percentage (MAPE) errors. After recovering from the initiation bias (as indicated by the vertical lines starting from level 0 on the y-axes), the fits produced by the two models emulate the timeseries reasonably well and partially capture the zero-valued dips caused by the Covid lockdowns. In the case of 'The Very Hungry Caterpillar' the forecast predicted agrees well with the last 32 weeks within two standard deviations, whereas in the case of 'The Alchemist' the forecast undershoots the last seasonal spike and the confidence interval resulting from the error is unable to compensate. The predicted sales volumes, per book title, are on average far off by roughly 20% compared to the actual sales records. For a well-established industry, a reasonable MAPE on a sales forecast typically falls between 10% and 30% [4].

The Very Hungry Caterpillar: SARIMAX (2, 1, 1) x (1, 0, [1, 2], 52)		The Alchemist: SARIMAX (1, 1, 2) x (2, 0, [ ], 52)	
Non-Seasonal Components	**Seasonal Components**	**Non-Seasonal Components**	**Seasonal Components**
AR(2): Uses 2 autoregressive lags, to regress the current value based on the previous 2 weeks.	**AR(1):** Uses 1 seasonal autoregressive lag to regress the current week's value on the value from exactly 52 weeks ago.	**AR(1):** Uses 1 autoregressive lags, to regress the current value based on the previous week.	**AR(2):** Uses 2 seasonal autoregressive lags to regress the current week's value on the value from 52 weeks and 104 weeks ago.
diff(1): Differences once to handle trend.	**diff(0):** No seasonal differencing is applied, meaning the seasonal pattern is assumed to be stable.	**diff(1):** Differences once to handle trend.	**diff(0):** No seasonal differencing is applied, meaning the seasonal pattern is assumed to be stable.
MA(1): Uses 1 forecast error to capture residual noise patterns which helps with short-term shock smoothing.	**MA([1,2]):** Includes the forecast errors from exactly 52 weeks ago (lag 1, L^m) and exactly 104 weeks ago (lag 2, L^{2m}).	**MA(2):** Uses 2 forecast errors from 1 and 2 weeks ago to capture residual noise patterns.	**MA(0):** No historical seasonal forecast error is used to adjust the current prediction.
No intercept.	**m=52:** Owing to weekly time series - 52 units per year.	An intercept is added.	**m=52:** Owing to weekly time series - 52 units per year.

Table 1: The best models returned by the auto ARIMA algorithm, described for 'The Very Hungry Caterpillar' (right) and 'The Alchemist' (left) case. Note that 'X' in SARIMAX signifies the model can include external variables

XGBoost Forecast Model

XGBoost is an ensemble technique that extends on traditional gradient boosting through the utilisation of regularisation and post-pruning elements. Considering XGBoost is a supervised learning algorithm, the pipeline was set-up with a recursive forecaster that generates a *sliding window* over the training set which looks back multiple time steps, to predict the next time step recursively.

A detrender (*PolynomialTrendForecaster* of degree 1) and an additive deseasonaliser were also included in the pipeline, where the full list of model parameters which has been selected post hyperparameter tuning is provided in Appendix C. To improve model performance against the Covid-19 zeros, Yeo-Johnsson transformation was performed to partially supress the variance while context was supplied in the form an exogenous variable (e.g. $'is_Covid' = 1\ or\ 0$). The forecasts obtained with the best performing XGBoost model is depicted in Figure 4.

LSTM Forecast Model

LSTM is an advanced type of Recurrent Neural Network (RNN) designed to overcome the vanishing/exploding gradient problem and handle long-range dependencies more effectively. Despite the more complex architecture, involving a *cell state* and three gates, LSTM networks maintain the recursiveness and require the data to be organised into 3D sequences of fixed length.

Therefore, for each title, the univariate timeseries was first normalised to the $(0,1)$ range and then, organised into 3D sequences of train-test sets. Unlike the XGBoost model, the LSTM networks were trained to forecast in a multi-step fashion with the lookback window length pre-tuned to overcome the Covid-19 distortion on the target scale as much as possible. The inference windows were further configured to avoid the Covid-19 zeros. Next, hyperparameter tuning was applied where the best performing model parameters are reported in Appendix D.

The results obtained are presented in Figure 5, where a significant general discrepancy between the actual and predicted sales volumes are evident for 'The Very Hungry Caterpillar'. The discrepancy in the amplitude of the last seasonal peak for 'The Alchemist' is in contrast much smaller. Although it is encouraging that the patterns predicted are reasonably structured, the zero values in the Covid-19 window cause the non-zero input values to be down-scaled. Note the effect is amplified for 'The Very Hungry Caterpillar' due to the timeseries possessing a higher variance. One way to tackle the performance issue observed for 'The Very Hungry Caterpillar' timeseries would be to experiment with different LSTM architectures.

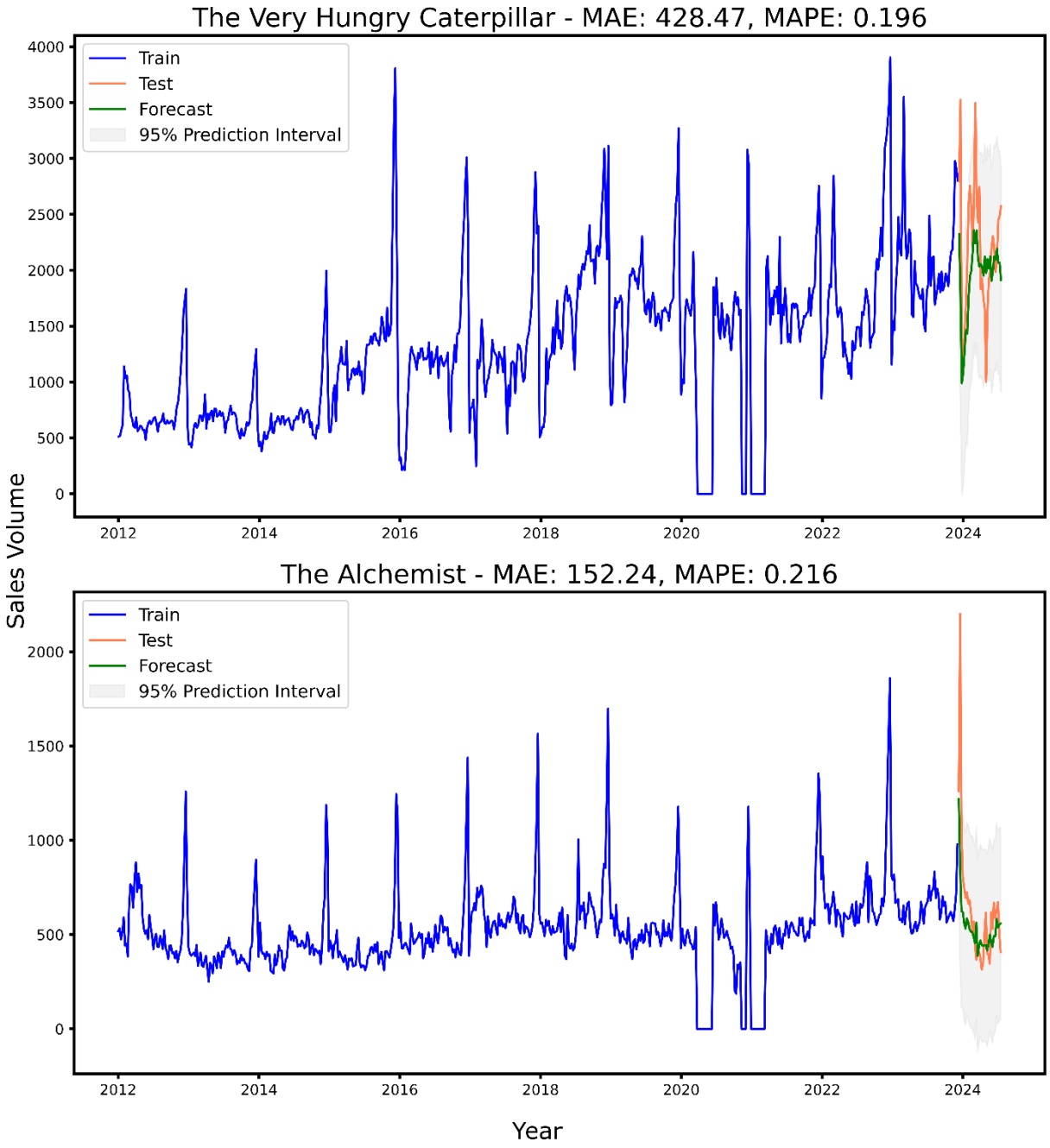


Figure 4: The XGBoost forecast obtained for 'The Very Hungry Caterpillar' (top) and 'The Alchemist' (bottom) timeseries, where the actual sales volumes are plotted in orange and the predicted values are plotted in green, with the shaded grey regions depicting the (95%) confidence intervals. Although, the MAE and MAPE scores are within an acceptable range (i.e. between 20-25%), the sales patterns predicted by the model deviate from the actual. In both cases, the model strongly relies on the seasonal pattern, which causes 'The Very Hungary Caterpillar' forecast to follow the spike around 2024 with a broader, shallower peak. 'The Alchemist' forecast on the other hand, follows the seasonal trend with a sharp peak followed by a shallower smaller peak correctly but significantly undershoots both the rise and the fall upon deviating away from the long-term trend.

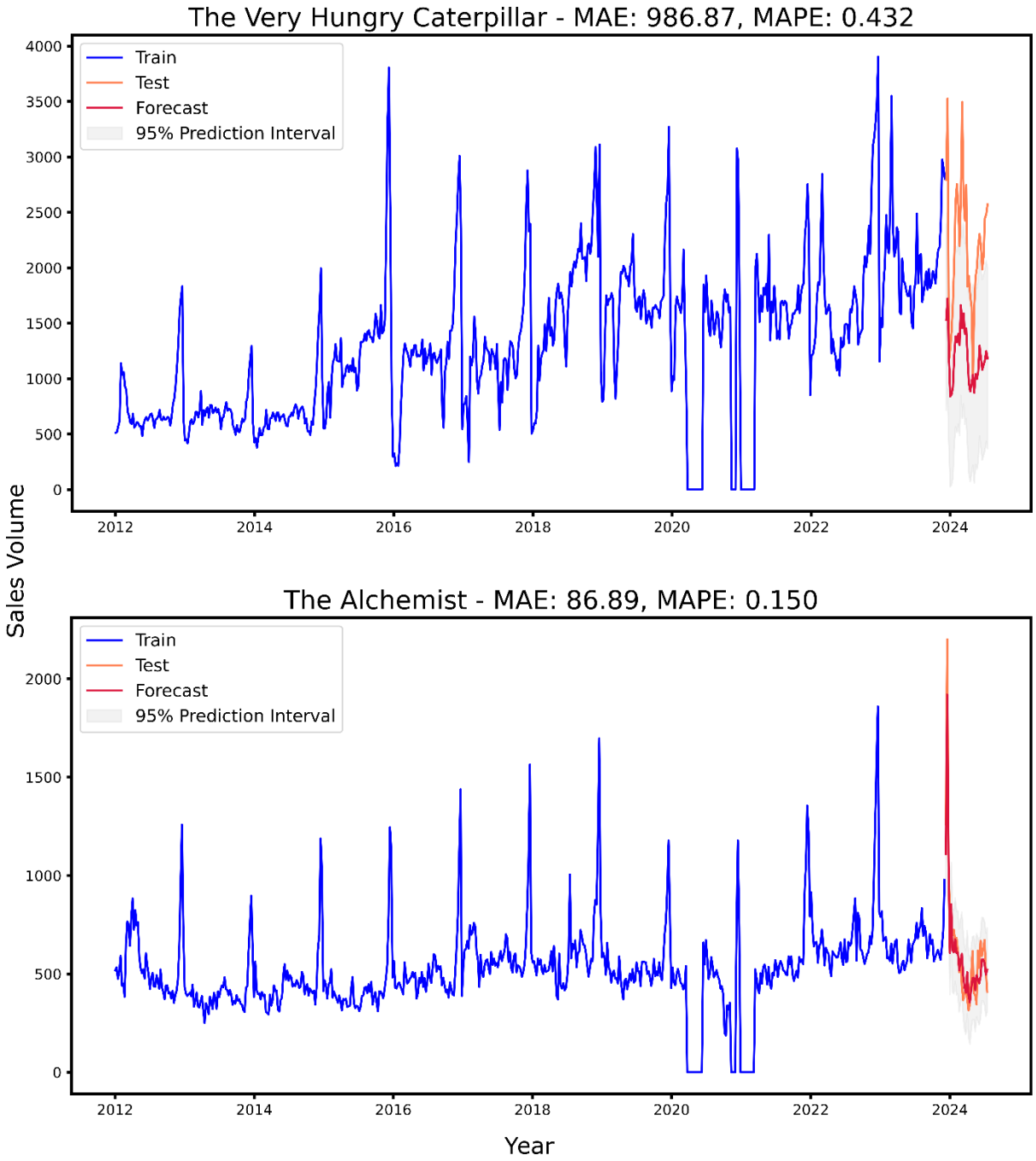


Figure 5: The LSTM forecast obtained for 'The Very Hungry Caterpillar' (top) and 'The Alchemist' (bottom) timeseries, where the actual sales volumes are plotted in orange and the predicted values are plotted in crimson, with the shaded grey regions depicting the (95%) confidence intervals. While the model predictions for 'The Alchemist' yield a good MAPE score as well as a relatively good fit of the seasonal pattern, the predicted amplitude for the final seasonal spike is slightly lower. In contrast, the model predictions for 'The Very Hungry Caterpillar' returns an extremely high MAPE score at 43.2% but broadly speaking, follows the sales pattern within the last 32 weeks. The large gap between the predicted and the actual values in the case of 'The Very Hungry Caterpillar' could stem from the scale distortion caused by the zero-valued gaps. While the effect is present for both datasets, 'The Very Hungry Caterpillar' sales volume has a larger variance than that of 'The Alchemist' range, which increases the impact of scaling further.

Hybrid Approaches

Two hybrid approaches combining SARIMA and LSTM models have been implemented:

1. A sequential combination wherein the residuals from the SARIMA model (Figure 6), were used to train the LSTM model and the final forecast was constructed from the sum of LSTM-predicted residuals (Figure 7) and the initial sales volume predictions made by SARIMA. The results are reported in Figure 8.

2. A parallel combination wherein the predictions from SARIMA and the predictions from LSTM were combined in the form of a weighted average, using 3 different pairs of weightages. The results obtained per title are reported in Figures 9 and 10.

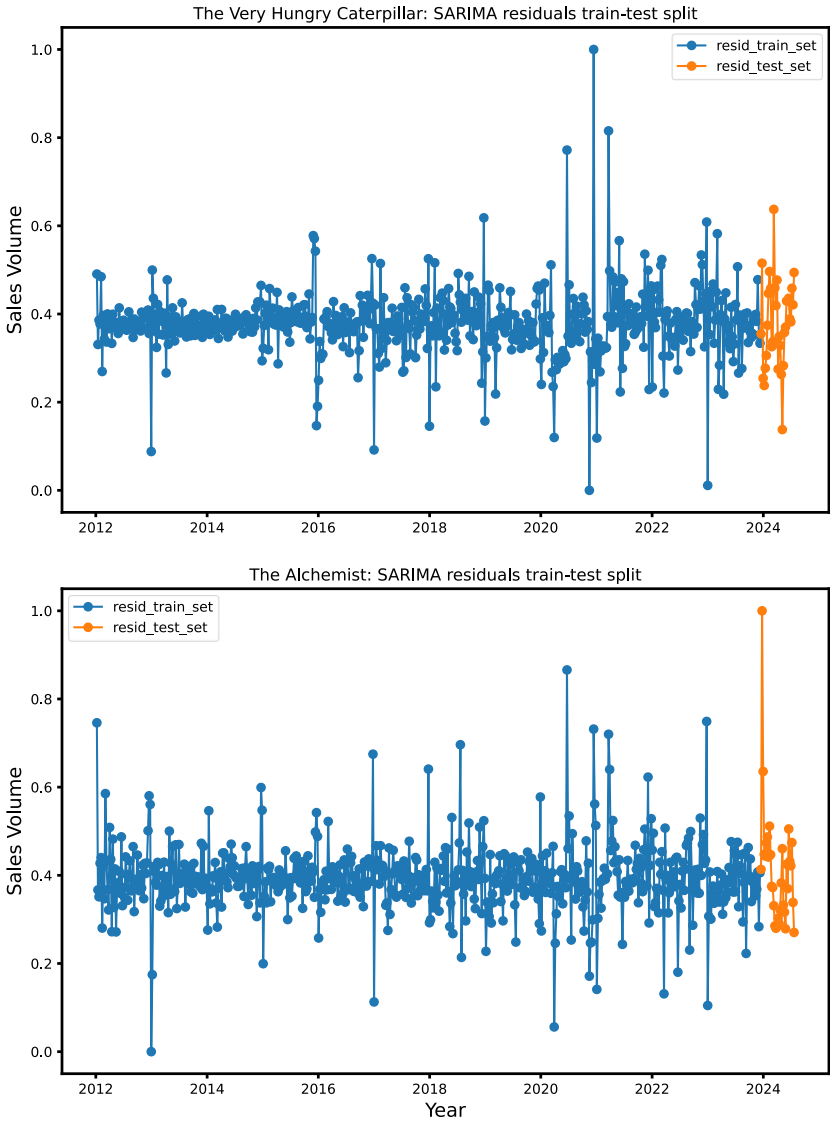


Figure 6: The residuals of the SARIMA fits (Figure 3) separated into train-test sets. The train set was further split using a rolling-window approach, producing multi-step 3D sequences. The LSTM models trained on these sequences was then used to predict the residuals of the final 32 weeks. In both cases, the residuals with the largest magnitudes occur within the Covid-19 window. The difference in the SARIMA-predicted and actual amplitude of the last seasonal peak within 'The Alchemist' timeseries is also reflected by the large jump at the start of 2024.

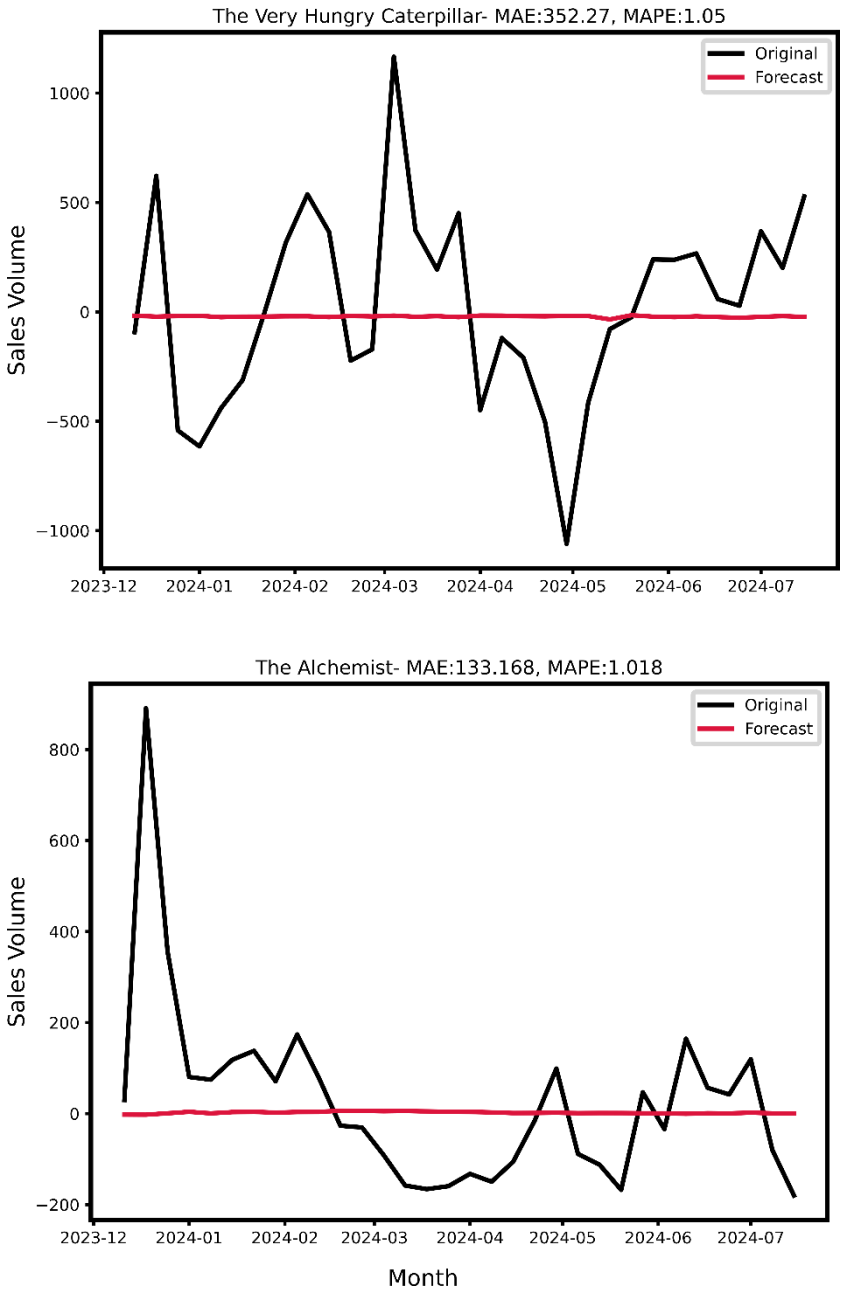


Figure 7: LSTM predictions made on the SARIMA residuals belonging to the final 32 weeks (test set in Figure 6. For both titles, the LSTM model fits a nearly flat line with a mean around zero, despite further hyperparameter tuning. Thus, little to no improvement can be expected for the sequentially combined hybrid forecast shown in Figure 8.

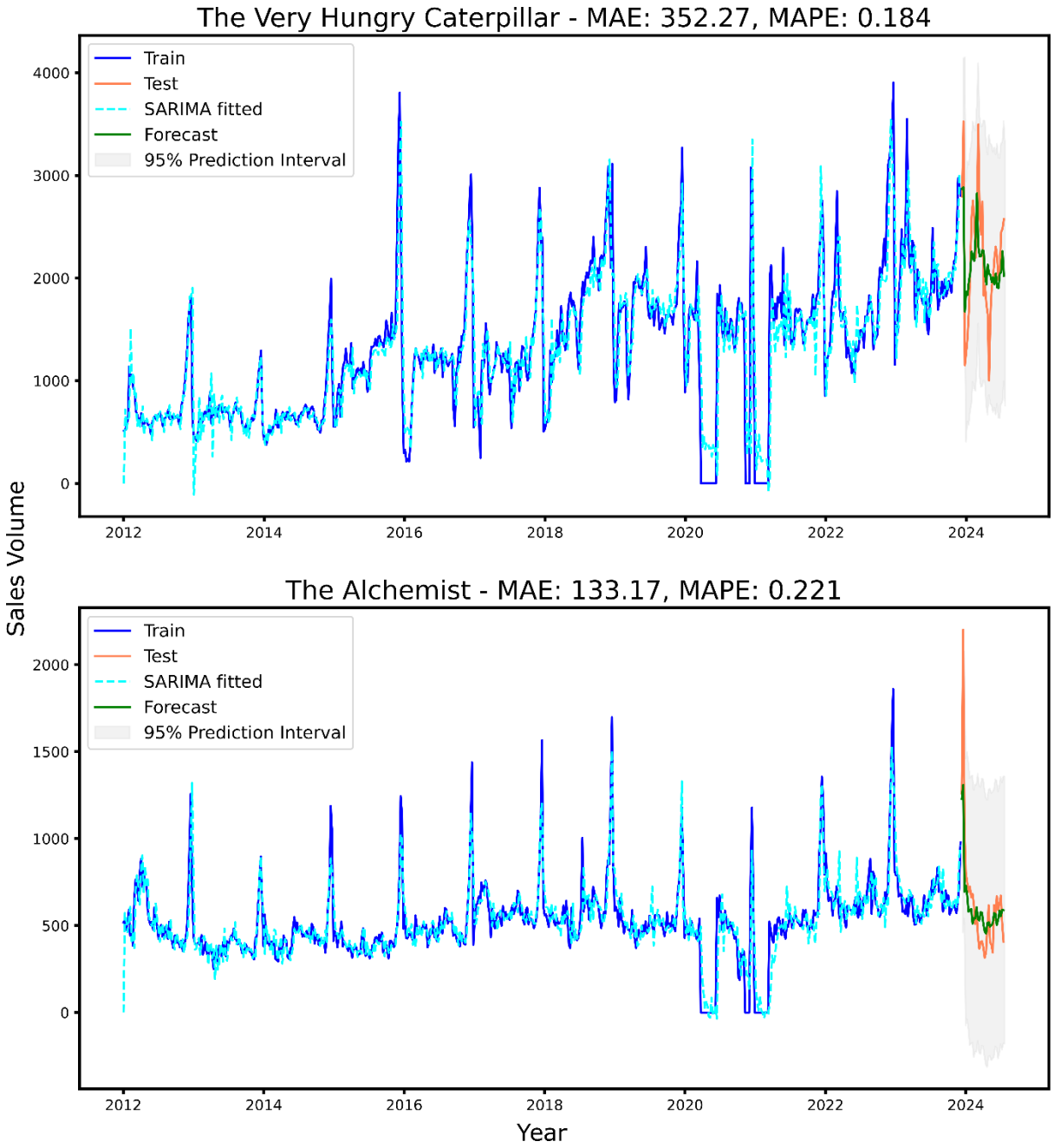


Figure 8: The sequentially combined hybrid forecasts obtained upon the addition of LSTM predicted residuals, depicted in Figure 7. Since the LSTM fits yield a mean around zero, the LTSM-predicted residuals have negligible effect on the error scores compared to the stand-alone SARIMA forecasts provided in Figure 3.

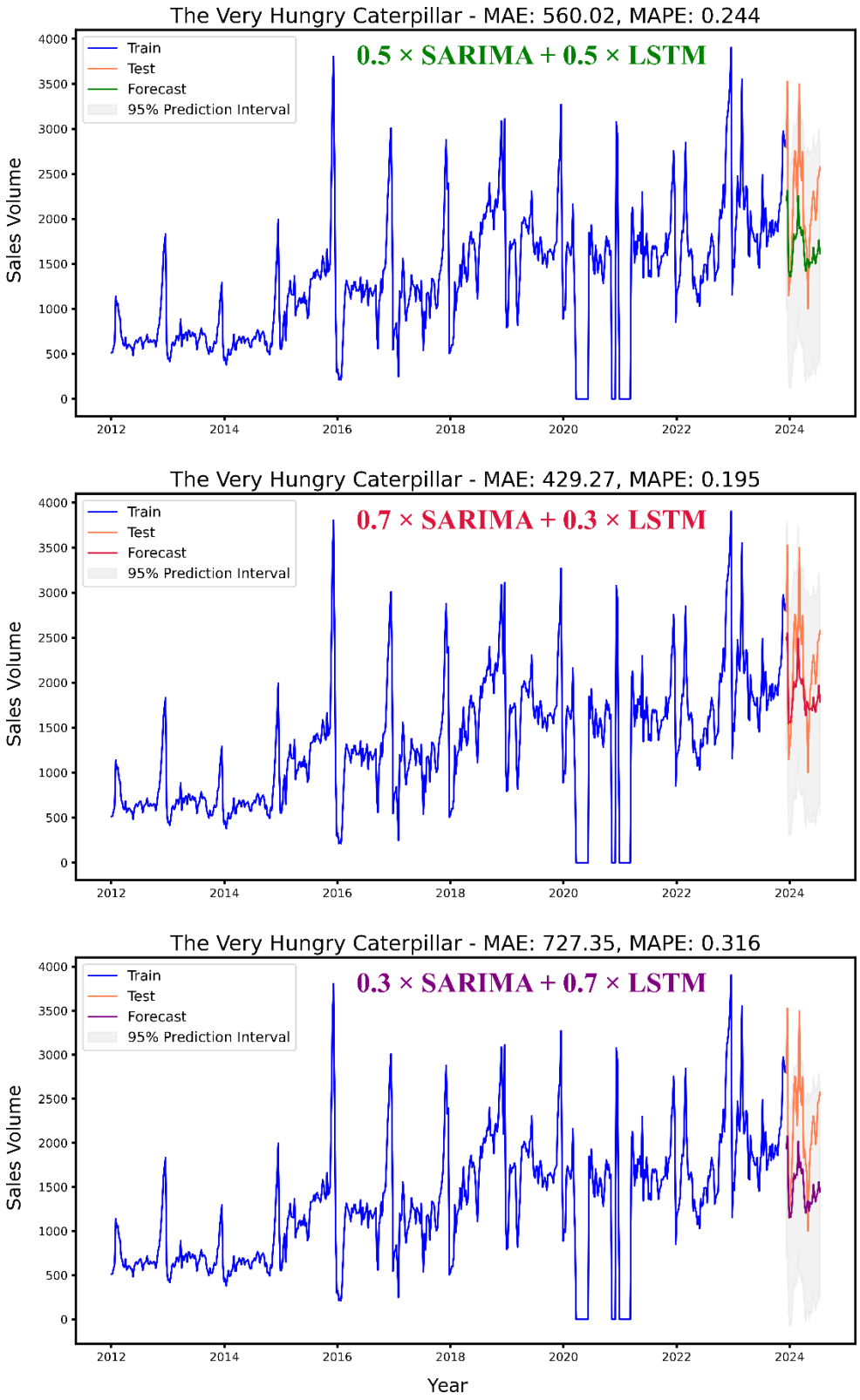


Figure 9 : Different weight combinations applied to the SARIMA+LSTM forecasts performed on 'The Very Hungry Caterpillar' timeseries. Since LSTM predictions suffer a scaling issue, the combination where the largest weight is given to the SARIMA model performs best but still fairs worse than the stand-alone SARIMA forecast depicted in Figure 3.

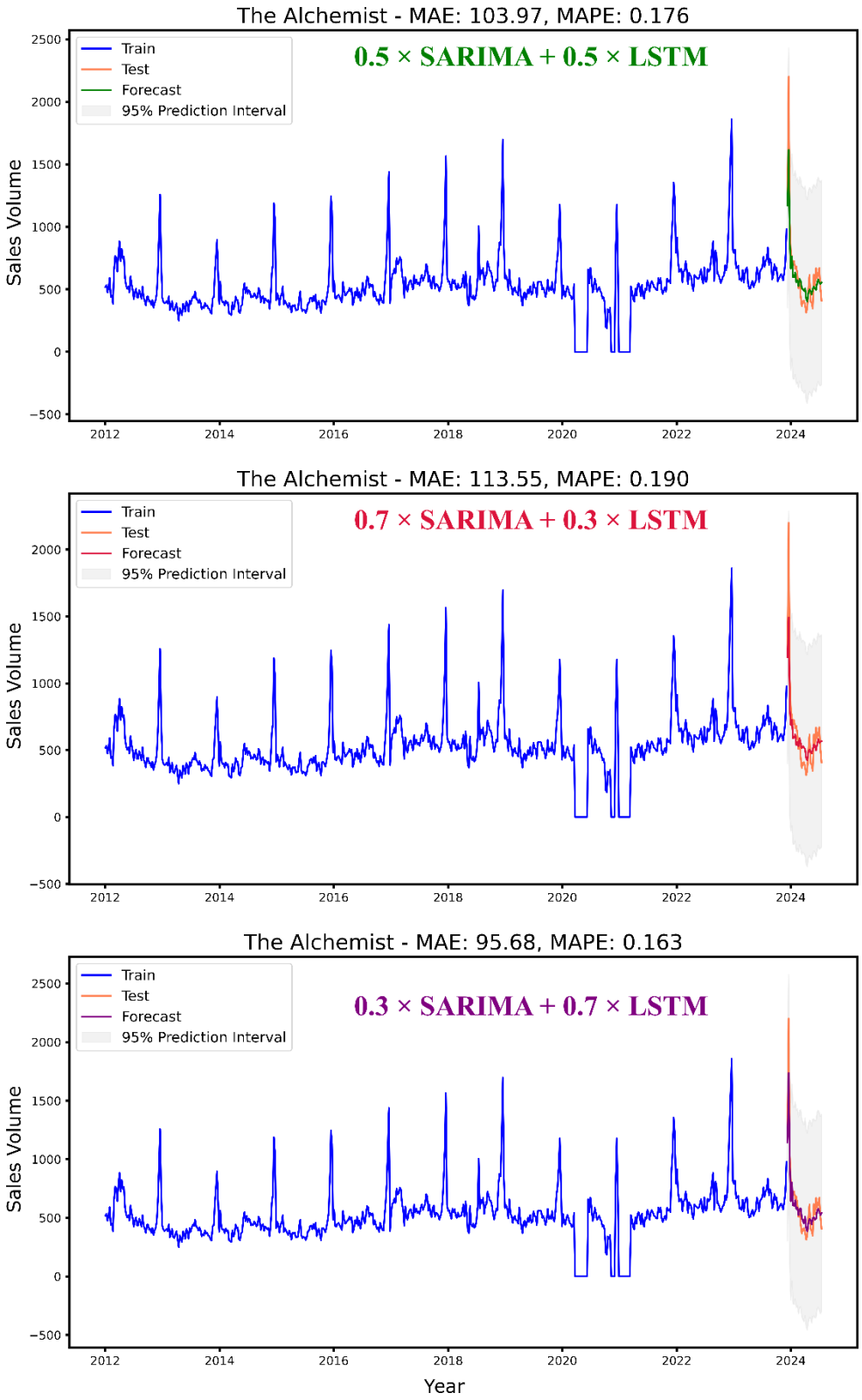


Figure 10 : Different weight combinations applied to the SARIMA+LSTM forecasts performed on 'The Alchemist' timeseries. Both models undershoot the amplitude of the last seasonal peak (around early 2024), but the LSTM model yields a better match. Thus, the winning combination is one where the LSTM forecast is given more weight. Note the stand-alone LSTM forecast depicted in Figure 5 still outperforms the best combination depicted here.

Using Monthly Sampling Frequency

The effect of increasing the sampling frequency was investigated using XGBoost and SARIMA. The XGBoost and SARIMA forecasts are provided in Figures 11 and 12 respectively , whereas the fitted SARIMA model components are described in Table 2.

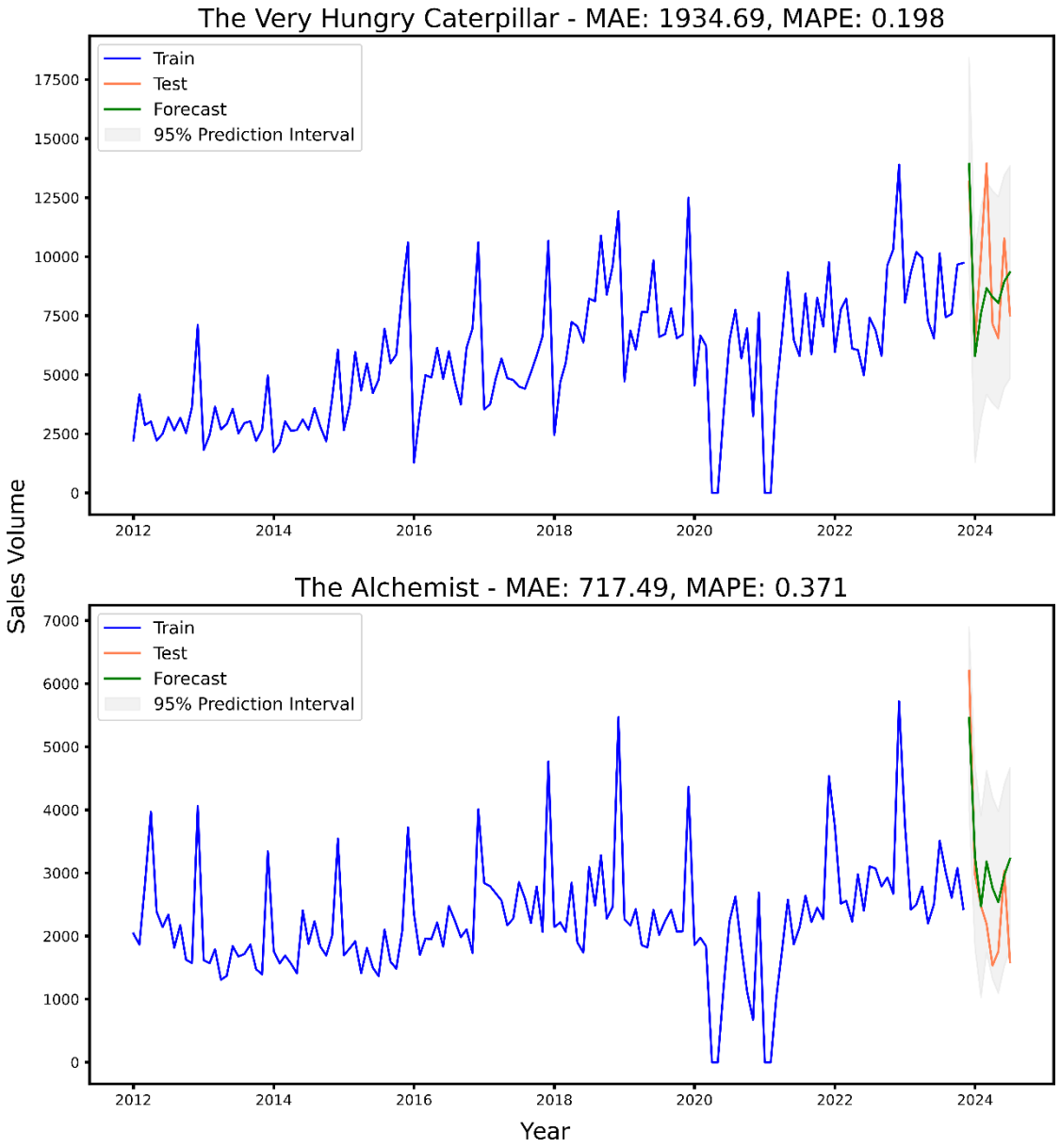


Figure 11 The XGBoost forecast obtained for the monthly timeseries, where the actual sales volumes are plotted in orange and the predicted values are plotted in green, with the shaded grey regions depicting the (95%) confidence intervals. Evidently, converting the weekly timeseries into monthly results in a 75% reduction in data points. While such a reduction helps absorb a fraction of the zero-valued Covid-19 gaps, the compressing of weeks into monthly bins also increases variance across the timeseries. For 'The Very Hungry Caterpillar' timeseries which presents with broader shorter peaks, the compressed distribution across the monthly grid helps close the gap between the actual monthly volume sales and the forecast predictions. Yet the opposite is true for 'The Alchemist' timeseries which presents with much sharper peaks and suffers from over-smoothing.

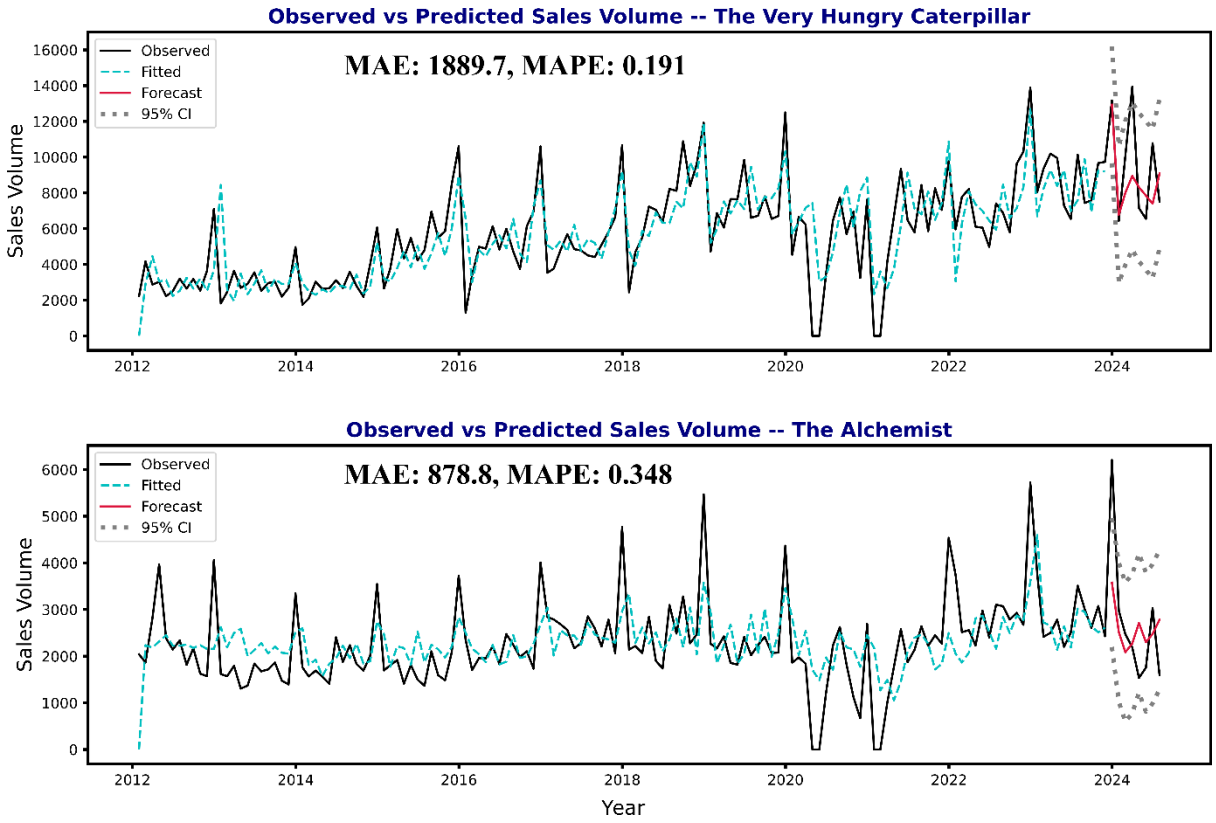


Figure 12 Upon converting to monthly data, the sampling grid becomes much coarser which causes the SARIMA models to struggle with the fits. For both timeseries, the 'burn' period where the model parameters suffer an initiation bias also lasts longer (24 months) compared to the weekly case. While the forecast produced for 'The Very Hungry Caterpillar' shows reasonable agreement with the actual sales volume within 2 standard deviations, the predictions obtained for 'The Alchemist' time series significantly undershoot both the final sharp peak and the dip that follows.

The Very Hungry Caterpillar: SARIMAX (1, 1, 1) x (1, 0, 1,12)		**The Alchemist : SARIMAX (0, 0, 1) x (1, 0, 1, 12)**	
AR(1): Uses 1 autoregressive lag, to regress the current value based on the previous month.	**AR(1):** Uses 1 seasonal autoregressive lag to regress the current month's value on the value from exactly 12 months ago.	**AR(0):** No autoregressive lags are used.	**AR(1):** Uses 1 seasonal autoregressive lag to regress the current month's value on the value from exactly 12 months ago.
diff(1): Differencing is applied once, to handle trend.	**diff(0):** No seasonal differencing is applied, meaning the seasonal pattern is assumed to be stable.	**diff(0):** No differencing is applied, meaning a stationary baseline is assumed.	**diff(0):** No seasonal differencing is applied, meaning the seasonal pattern is assumed to be stable.
MA(1): Uses 1 forecast error to capture residual noise patterns which helps with short-term shock smoothing.	**MA(1):** Includes one forecast error from 12 months ago (lag 1, L^m) .	**MA(1):** Uses 1 forecast error to capture residual noise patterns which helps with short-term shock smoothing.	**MA(1):** Includes one forecast error from 12 months ago (lag 1, L^m) .
No intercept.	**m=12:** Owing to monthly time series - 12 units per year.	An intercept is added.	**m=12:** Owing to monthly time series - 12 units per year.

*Table 2 : The SARIMAX models returned by the auto ARIMA algorithm using **monthly** sampling. Note that 'X' in SARIMAX signifies the model can include external variables*

Conclusions:

The following table highlights the key results obtained for the weekly time series.

Model	The Very Hungry Caterpillar MPE Scores	The Alchemist MPE Scores	15-25% Error Range	
SARIMA	0.185	0.219	✓	✓
XGBoost	0.196	0.216	✓	✓
LSTM	0.432	0.150	×	✓
SARIMA-LSTM Sequential	0.184	0.221	✓	✓
SARIMA-LSTM Parallel*	0.195	0.163	✓	✓

*Table 3: Summary of key metrics obtained for the weekly timeseries, where * indicates that the best performing weight combinations have been quoted for the parallel-wise SARIMA-LSTM hybrid models. Note that the acceptable percentage error range is assumed to be below 25% for the book retail sector.*

Based both on the error scores and the resulting forecast patterns, the SARIMA and XGBoost models prove more robust against the presence of zero-valued data points but fail to truly capture the long-term trends due to the Covid-19 discontinuities in the sales volume.

To help absorb the short-term shock which the forecasted errors cannot accommodate, context was provided to the XGBoost models in the form an exogenous variable (e.g. $'is_Covid' = 1 \; or \; 0$). To help stabilise the variance (exacerbated by the Covid-19 zeros) without significantly altering the sales records, a Yeo-Johnsson transformation was performed. The two approaches in combination help boost the regressor performance but should be further supported by an effective tuning approach that involves an extended hyperparameter space.

While an XGBoost regressor can manage 27 weeks of zero sales data, the long-term LSTM memory can become significantly distorted, for it is constructed through a series of 'tanh' and 'sigmoid' activations at the gates. Therefore, a different inference strategy was adopted whereby the sliding windows coinciding with the Covid-19 event windows (i.e., lockdown periods) were completely disregarded. Still, the distortion in the scaled input range proved problematic for 'The Very Hungry Caterpillar' timeseries which may benefit from a different LSTM architecture [5–9] .

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Appendix A

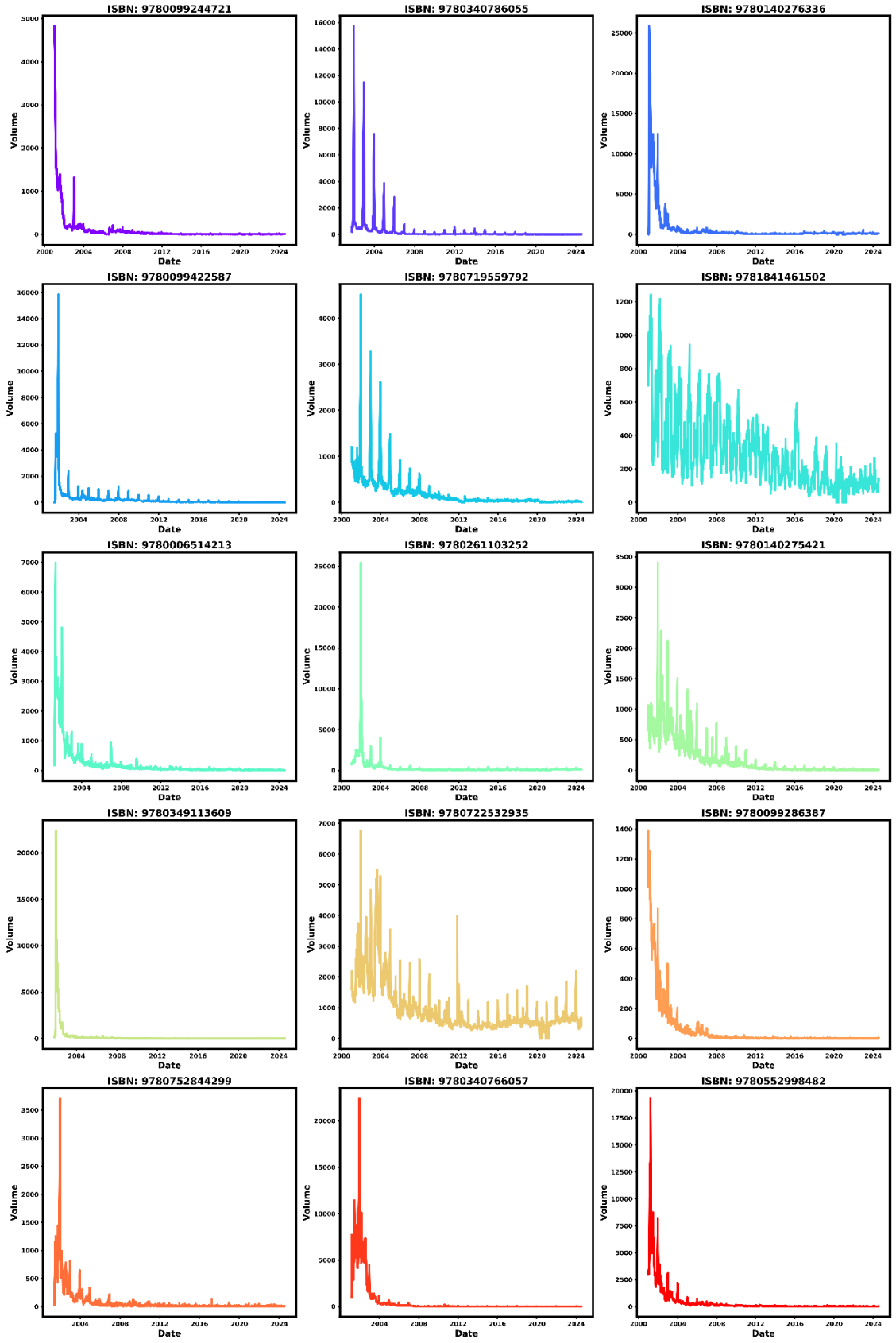


Figure A: A subset of the historical sales data produced for 15 of the 61 unique ISBNs with data beyond 2024-07-01.

ISBN	Title
9780099244721	Timeline
9780340786055	One Hundred Ways for a Cat to Train Its Human
9780140276336	White Teeth
9780099422587	London: The Biography
9780719559792	Now We Are Sixty
9781841461502	KS2 English Study Book - Ages 7-11
9780006514213	Miss Garnet's Angel
9780261103252	Lord of the Rings
9780140275421	The Other Side of the Dale
9780349113609	Last Time They Met
9780722532935	The Alchemist
9780099286387	Geisha
9780752844299	Arthur: The Seeing Stone
9780340766057	McCarthy's Bar
9780552998482	Chocolat

Table A: For reference, a list of the titles associated with the ISBNs whose timeseries have been plotted in Figure A.

Almost every title experiences a sharp rise starting from 2001 which is when Nielsen BookScan starts tracking consumer point-of-sale data for the book/publishing sector, followed by some oscillations in sales volume and finally, a continuous decline with the overall trend resembling an exponential decay.

Despite the difficulty in visually assessing a set of timeseries spanning over 23 years and exhibiting high variance, the distribution of peaks/spikes across most of the weekly sampled grids suggests a seasonal element.

To understand the general sales patterns of the 15 titles represented in the plots, it is important to avoid speculating on the potential reasons behind the popularity of a particular title or genre but instead focus the discussion on broad trends dictating consumer behaviour. For instance, the year 2008 coincides with the Great Recession and the year 2007 marks the release of the first Kindle e-reader.

It is also well known that the technological advancements such as widespread use of smart phones and the availability of faster broadband helped trigger a massive rise for the e-commerce sector and instability for the high-street retail, starting from 2000s and continuing through to 2010s. The publishing sector was one of the first affected whereby consumers no longer had to purchase books in print and had several other options including various e-book and audio-book platforms.

This period of constant change is reflected in the book sales data where the massive fluctuations start to dampen typically within the 2004-2008 window. Thus, for most book titles the period covering the 12-24 years appear more stable compared to the period 1-12 years.

Mathematically speaking, the high variance and the exponential behaviour of the sales data would likely require some form of truncation function to stabilise the models implemented in the current study.

Appendix B

The following table lists several test statistics returned for the residuals obtained from the SARIMA model fits performed on 'The Very Hungry Caterpillar' and 'The Alchemist' timeseries, depicted in Figure 3.

Statistic	The Very Hungry Caterpillar
Prob(Q)	0.94 (> 0.05), no significant autocorrelation remains
Prob (JB)	0.00 (< 0.05), rejects normal distribution
Prob (H)	0.00 (< 0.05), rejects constant variance
Statistic	**The Alchemist**
Prob(Q)	0.97 (> 0.05), no significant autocorrelation remains
Prob (JB)	0.00 (< 0.05), rejects normal distribution
Prob (H)	0.00 (< 0.05), rejects constant variance

Table B: Residual diagnostics obtained from the SARIMA model fits, where Prob(Q) refers to the Ljung-Box Q (LBQ) statistic with the null hypothesis stating no autocorrelation remains; Prob(JB) refers to the Jarque-Bera statistic with the null hypothesis stating the distribution is normal; Prob(H) refers to the heteroskedasticity test with the null hypothesis stating the variance is constant (no heteroskedasticity).

Based on the residual statistics presented above and the distributions presented in Figure B, the residuals appear randomly distributed where the distributions could be described as nearly normal. However, the non-constant variance suggests an element of non-stationarity, implying the SARIMA models do not fully account for the structure present in the weekly timeseries. At the same time, the Covid-19 event window could be a major contributor of the discrepancy, magnitude, and influence of which is reflected in the residual plots in Figure 6.

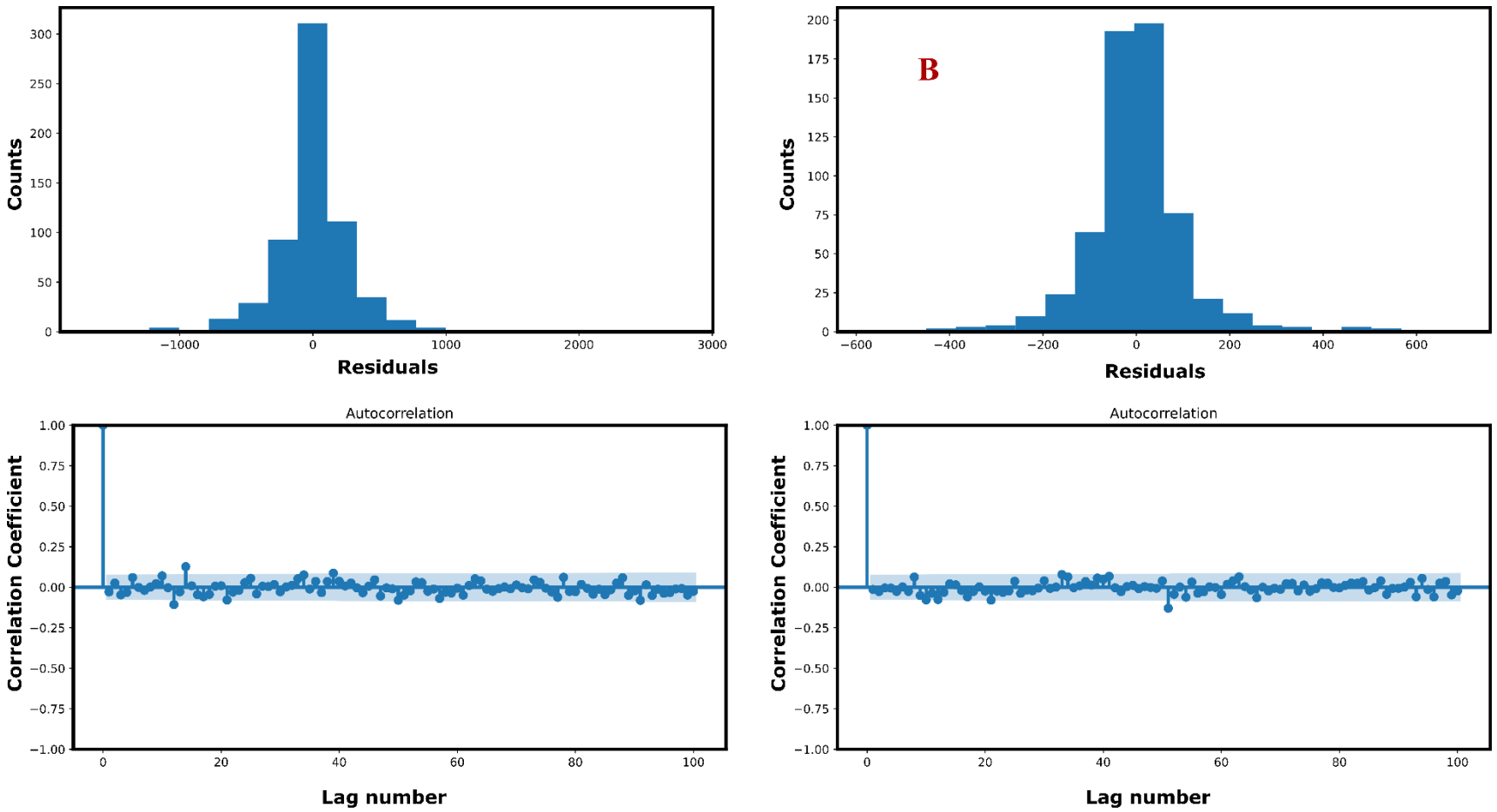


Figure B: The residuals resulting from the SARIMA-fitted weekly timeseries data, with 'The Very Hungry Caterpillar' residual distribution and Autocorrelation Function (ACF) plots shown on the left (A) and 'The Alchemist' residual distribution and ACF plots shown on the right (B). In both cases, the central portion of the residual distribution resembles a normal distribution with a mean around zero as would be in the case of white noise. However, a heavy tail is also present with a large standard deviation signifying high variance. When focussing on the ACF plots, most of the lags fall within the region of insignificance, except lags 12 and 14 in 'The Very Hungry Caterpillar' case and lag 52 in 'The Alchemist' case which slightly stick out. With both observations considered, the ACF plots suggest no structural pattern remains and that the residuals are likely random.

Appendix C

The best model was determined by performing a grid search with 11-fold cross-validation over a predetermined hyperparameter space. The set of parameters which resulted in the lowest MAE and MAPE scores as well as the best fits are listed in table below.

XGB Regressor Hyperparameters	Value Applied
n_estimators	400
min_child_weight	1
max_depth	7
learning_rate	0.1
reg_alpha	0
colsample_bylevel	1
colsample_bynode	1
colsample_bytree	1
gamma	0
subsample	0.5
forecaster window_length	52

Table C: The optimal set of hyperparameters that displayed the best performance against other combinations tried, following a grid search.

Appendix D

The best model was determined using a random search over a predetermined hyperparameter space, following a set of manual tests on the optimal lookback window length. The parameter combinations resulting in the best error scores as well as best fits to the test data are listed in the table below.

Hyperparameter	The Very Hungry Caterpillar	The Alchemist
Nodes (layer 1)	192	160
Dropout (layer 1)	0.4	0.3
Nodes (layer 2)	160	160
Dropout (layer 2)	0.0	0.0
Output (dense layer)	32	32
Activation (dense layer)	ReLU	ReLU
Learning Rate	0.01	0.01
Lookback Window Length	52	78

Table D: The optimal set of parameters obtained for the two LSTM models, upon tuning using a random search.

The stacked LSTM architecture operates as follows:

- **First LSTM layer:** Processes the 3D input sequence and returns the hidden state for every time step in the lookback window. Thus, it passes a 3D sequence to next layer.
- **Second LSTM layer :** Processes the sequence from the first layer and outputs only the final hidden state vector after processing the entire sequence. In other words, it compresses the temporal data into a 2D tensor.
- **Dropout layers:** The dropout layers help with regularisation by randomly deactivating a given percentage of the neurons during training to prevent overfitting.
- **Output layer:** Maps the final LSTM hidden state directly to the output vector of size (1,32).